Exercises: Building A Small Dataset of Economists

Data Analytics with Python for Economics

Choose **two or three economists** you find inspiring. You may choose any economists. You can look up biographical information online if necessary.

Work through the tasks in order. **Tasks 1 and 2 are the main exercise.** Continue with the optional tasks if you have time.

Recommended: Work in a Jupyter Notebook (.ipynb) so you can execute and inspect your code step by step.

1. Lists and tuples

1. Create a list called `an_economist` containing the **name** and **year of birth** of your first economist.
2. Use `.append()` to add the **year of death**, **country of birth**, and **Nobel Prize year**, in that order.
3. If your economist is alive or did not receive the Nobel Prize, use `False` for the corresponding year.
4. Display the list. Change one entry using its index. Remember that Python starts counting at `0`.
5. Convert your list to a tuple. What happens if you try to change an entry now?

Starting point:

```
an_economist = ['Your chosen economist', 1900]  # Replace both values
# Continue here
```

2. Dictionaries

Create a dictionary for your **second economist** using these five keys: `'name'`, `'year_of_birth'`, `'year_of_death'`, `'country_of_birth'`, `'nobel_prize_year'`

Then:

1. Display the dictionary.
2. Access the economist's name using the key `'name'`.
3. Correct or change one value by referring to its key.
4. Why might a dictionary be easier to work with than a list containing the same information?

Starting point:

```python
another_economist = {
    'name': 'Your second economist',
    'year_of_birth': 1900,
    'year_of_death': False,
    'country_of_birth': 'Country',
    'nobel_prize_year': False
}
# Replace the example values and continue here
```

3. Optional: Build a small table with pandas

Create a **third economist** as a dictionary with the same five keys. Put your dictionaries into a list called `some_economists`. (You may also convert your first economist's tuple into a dictionary.)

Then create a pandas `DataFrame`:

```python
import pandas as pd

some_economists = [another_economist]  # Add your other dictionaries
some_economists_df = pd.DataFrame(some_economists)

print(some_economists_df)
print(some_economists_df.dtypes)
```

Which columns represent years? Are their data types suitable?

4. Optional: Missing values

Use the pandas methods `astype()`, `convert_dtypes()`, and `replace()` to prepare the columns containing years. Replace `0` and `False` where they indicate missing information with `pd.NA`, and ensure that the year columns have suitable data types that allow missing values.

Check: Display your final `DataFrame` and its `.dtypes`.